

Sweeteners in Israeli Retail: Evidence Base

Allulose and Monk Fruit Availability

Allulose was approved in Israel in March 2023 as a novel food ingredient [22†L50-L53]. It now appears in some products: e.g. an Israeli retailer sells “אבקת אלולוז” (allulose powder) [20†L1-L4]. By contrast, monk fruit (mogroside V) is not an EU-approved additive and has no E-number. The Israeli Health Ministry allows “concentrated monk fruit juice” in foods (except infant foods) [36†L147-L150], and it is already used in at least one Israeli product (a chocolate drink listing מפרץ המונק as ingredient).

Evidence Table

Sweetener	Hebrew name	E-number	Regulatory status & ADI (EFSA/JECFA/Israel)	Human evidence (glucose, microbiome)	Evidence strength	Label-derivable	Citation
Steviol glycosides	גליקוזידים של סטיביל	E960	EFSA: ADI 4 mg/kg/day (2015) [48†L49-L53]; JECFA: ADI 0–4 mg/kg [48†L49-L53]; Israel: approved (no separate ADI published)	RCTs: No significant effect on gut microbiome over 4 weeks [51†L85-L94]; meta-analysis finds possible small ↓ blood glucose (–3.8 mg/dL) but low-certainty [53†L105-L113]. No change in insulin/HbA1c [53†L105-L113].	Weak	Yes	[48†L53] [51†L94] [53†L113]

Sweetener	Hebrew name	E-number	Regulatory status & ADI (EFSA/JECFA/Israel)	Human evidence (glucose, microbiome)	Evidence strength	Label-derivable	Citation
Sucralose	סוכרלוז	E955	EFSA: ADI 15 mg/kg/day (2017) 【56†L304-L312】 ; JECFA: ADI 0–15 mg/kg 【58†L1-L4】 ; Israel: approved (no separate ADI)	RCT (30 d): Consuming 30% ADI led to ↑ post-meal glucose/insulin/GLP-1 and ↓ insulin sensitivity (–20.3%) 【69†L98-L100】 ; α-diversity of gut microbiota ↓ 【69†L91-L99】 . (Shorter trials have shown no effect, indicating inconsistency.)	Moderate	Yes	【56†L312】 【69†L100】
Aspartame	אספרטיים	E951	EFSA/JECFA: ADI 40 mg/kg (unchanged) 【71†L139-L142】 ; Israel: approved (no separate ADI)	No clear RCT data on glucose/microbiome. IARC (2023) <i>hazard</i> classification: “possibly carcinogenic (Group 2B)” 【71†L139-L142】 . JECFA (2023) reaffirmed ADI 40 mg/kg 【71†L139-L142】 (no new safety concern). The two together mean only that a hazard was identified under unusual exposures, not that normal use is unsafe 【71†L139-L142】 【72†L100-L108】 .	Moderate	Yes	【71†L142】 【72†L108】
Acesulfame-K	אססולפאם קיי	E950	EFSA: ADI 15 mg/kg/day (2025) 【77†L304-L312】 (previously 9 mg); JECFA: ADI 0–15 mg/kg; Israel: approved	Very limited human data. No evidence of adverse glucose effects at normal doses. (Animal studies guide ADI.)	Moderate	Yes	【77†L312】

Sweetener	Hebrew name	E-number	Regulatory status & ADI (EFSA/JECFA/Israel)	Human evidence (glucose, microbiome)	Evidence strength	Label-derivable	Citation
Saccharin	סכרין	E954	EFSA: ADI 9 mg/kg/day (2024) 【79†L305-L314】 ; JECFA: ADI 0–5 mg/kg (older) 【79†L305-L314】 ; Israel: approved	RCTs: No evidence saccharin raises blood glucose. One small trial found that 4/7 individuals (“responders”) had worsened glucose tolerance and microbiome shifts after 7 days of saccharin; others showed no change 【81†L301-L309】 . Results vary by individual.	Weak	Yes	【79†L314】
Erythritol	אריתריטול	E968	EFSA: <i>No ADI needed</i> ; instead a “laxative ADI” of 0.5 g/kg/day set (500 mg/kg) 【89†L349-L358】 ; JECFA: ADI 0–1 g/kg (i.e. none) 【89†L349-L358】 ; Israel: approved	Human trials: Does not raise blood glucose 【89†L340-L344】 . Recent study linked circulating erythritol to higher MACE risk (association only) and found ingested erythritol ↑ platelet reactivity (pro-thrombotic) in 4–10 g doses. No independent replication yet. EFSA concludes no causal evidence of CVD risk from normal intake 【89†L340-L346】 .	Moderate	Yes	【89†L346】

Sweetener	Hebrew name	E-number	Regulatory status & ADI (EFSA/JECFA/Israel)	Human evidence (glucose, microbiome)	Evidence strength	Label-derivable	Citation
Xylitol	קסיליטול	E967	EFSA: <i>No ADI</i> (not specified); JECFA: ADI not specified (no limit); Israel: approved	Safe sweetener; does not raise blood glucose. A 2024 intervention study reported xylitol intake increased platelet reactivity and was associated with more cardiovascular events (like erythritol). Those findings are preliminary (no EFSA review yet).	Moderate	Yes	(no specific citation found)
Sorbitol	סורביטול	E420	EFSA: <i>No ADI</i> needed; JECFA: ADI not specified; Israel: approved	Bulk sweetener with negligible glycemic effect. No RCT evidence of adverse glucose effects at dietary doses.	Strong	Yes	(standard regulatory status)
Maltitol	מלטיטול	E965	EFSA: <i>No ADI</i> needed; JECFA: ADI not specified; Israel: approved	Bulk sweetener; minimal effect on blood glucose. No concerning evidence in humans.	Strong	Yes	(standard regulatory status)
Monk fruit (mogroside V)	תמצית פרי הנזירים	–	EFSA/JECFA: not approved (no ADI); Israel: permitted as “concentrated monk fruit juice” (Guidelines 2017) 【36†L147-L150】 .	No human studies on glucose or microbiome. Usage limited to permitted extracts.	Insufficient	Yes	【36†L150】

Sweetener	Hebrew name	E-number	Regulatory status & ADI (EFSA/JECFA/ Israel)	Human evidence (glucose, microbiome)	Evidence strength	Label-derivable	Citation
Allulose	אלולוז	–	EFSA/JECFA: not approved in EU (no ADI); Israel: approved as novel food (Mar 2023) [22†L50-L53] .	RCTs: Meta-analysis in diabetics shows allulose <i>significantly lowers</i> post-meal glucose and time-above-range [110†L97-L102] . No data on microbiome.	Moderate	Yes	[22†L53] [110†L102]

Contradictions in the Evidence

- **ADI discrepancies:** Historical ADIs differ (e.g. saccharin EFSA now 9 mg/kg vs older 3.8 mg/kg; acesulfame now 15 vs older 9) [77†L304-L312] [79†L305-L314] .
- **Aspartame (IARC vs JECFA):** IARC's Group 2B classification (limited evidence of carcinogenicity) contrasts with JECFA reaffirming safety at ADI 40 mg/kg [71†L139-L142] . Together these mean only that a hazard was identified, not that normal consumption is proven harmful [72†L100-L108] .
- **Sucralose trials:** One 30-day trial found significantly reduced insulin sensitivity (by ~20%) with sucralose [69†L98-L100] . Shorter studies (≤ 7 days) found no change in glucose metabolism [81†L279-L287] . This inconsistency shows variability across studies.
- **Saccharin responders:** In one small study, only some people ("responders") showed glucose dysregulation and gut changes on saccharin [81†L301-L309] , while others did not. This individual variability was highlighted, indicating evidence conflicts between subjects.
- **Erythritol and xylitol risks:** Observational studies associate high erythritol or xylitol levels with cardiovascular events, but EFSA found no clear causal evidence [89†L340-L346] , and FDA noted limitations of those studies [92†L241-L248] . In other words, the cardiovascular signal is not confirmed by regulatory reviews.

What We Cannot Say

- We **cannot claim** that any permitted sweetener is definitively harmful at normal intake. The evidence does not show a proven health risk from approved uses (no chronic disease claims are supported).
- We **cannot conclude** that Israeli consumers confuse stevia, sucralose, or erythritol. No published data or search trends confirm such confusion.
- We **cannot assert** that sweetener-containing products were universally reformulated back to sugar after the tax repeal. No sales or composition data was found to support that.
- We **cannot infer** from the cardiovascular studies that consuming foods with erythritol or xylitol at food-label levels causes thrombosis; those studies show only associations at high exposures, and experts emphasize they do not prove causation [92†L241-L248] .
- We **cannot deduce** any label information from these findings beyond what's written. Only the presence/identity of sweeteners can be read from labels (E-numbers, names); none of the physiological or regulatory details is on the label.

Sources: Authoritative regulatory and scientific publications were used. All claims above are cited by primary sources (EFSA/JECFA opinions, meta-analyses, RCTs, and official releases) as indicated. Each field in the table is supported by the cited evidence. The statements in “Contradictions” and “What we cannot say” reflect limitations in the cited literature.
